

EXHIBIT 1



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Ramani et al.

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(54) **METHOD TO REDUCE MOL DAMAGE ON NISI**

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H01L 29/78 (2006.01)

(52) **U.S. Cl.** **257/413**; 257/383; 257/E29.156

(58) **Field of Classification Search** 257/369, 257/383, 412, E29.139, E29.242, 382, 388, 257/413, E29.156, 377

See application file for complete search history.

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Primary Examiner — Matthew Landau

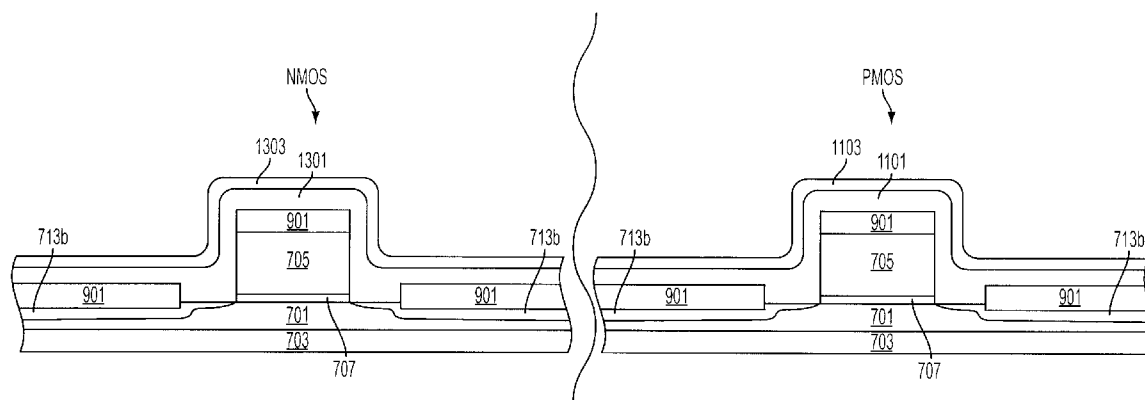
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(57) **ABSTRACT**

Transistor devices are formed with nickel silicide layers formulated to prevent degradation upon removal of overlying stress liners. Embodiments include transistors with nickel silicide layers having a platinum composition gradient increasing in platinum content toward the upper surfaces thereof, i.e., increasing in platinum in a direction away from the gate electrode and source/drain regions. Embodiments include forming a first layer of nickel having a first amount of platinum and forming, on the first layer of nickel, a second layer of nickel having a second amount of platinum, the second weight percent of platinum being greater than the first weight percent. The layers of nickel are then annealed to form a nickel silicide layer having the platinum composition gradient increasing in platinum toward the upper surface. The platinum concentration gradient protects the nickel silicide layer during subsequent processing, as during etching to remove overlying stress liners, thereby avoiding a decrease in device performance.

20 Claims, 13 Drawing Sheets



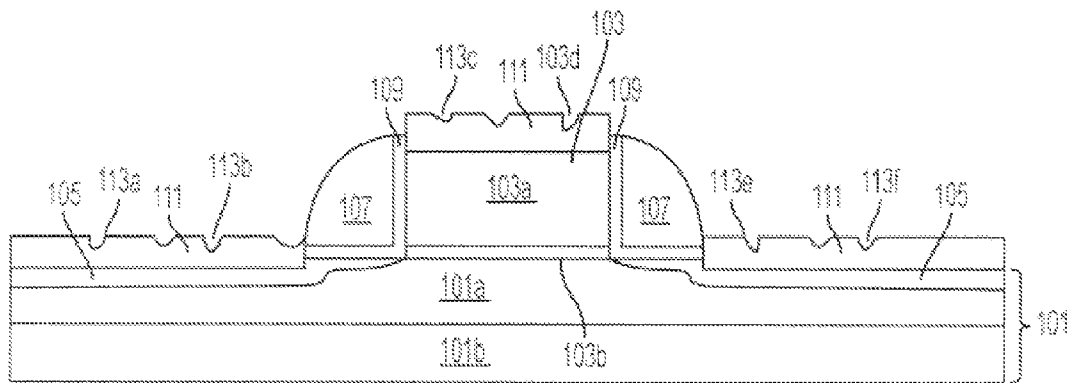


FIG. 1
(PRIOR ART)

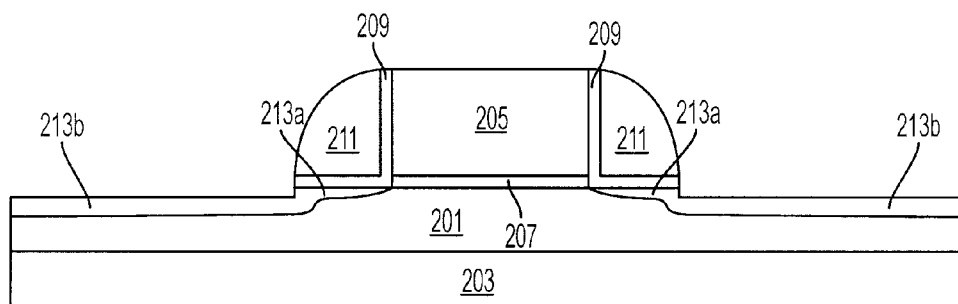


FIG. 2

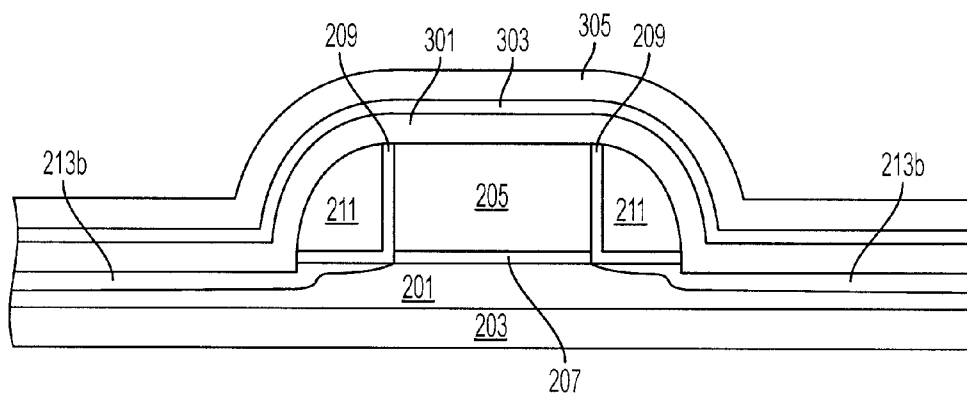


FIG. 3

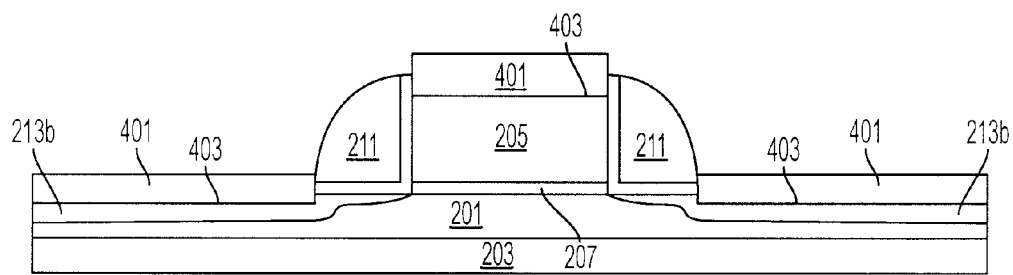


FIG. 4

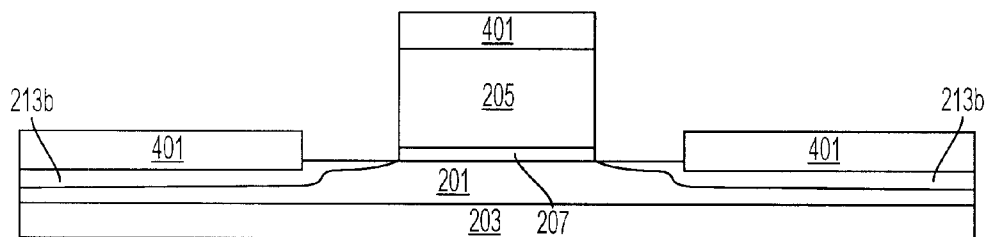


FIG. 5

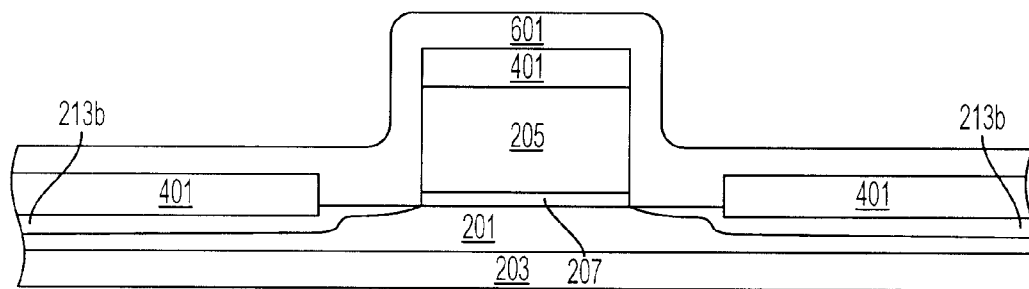


FIG. 6

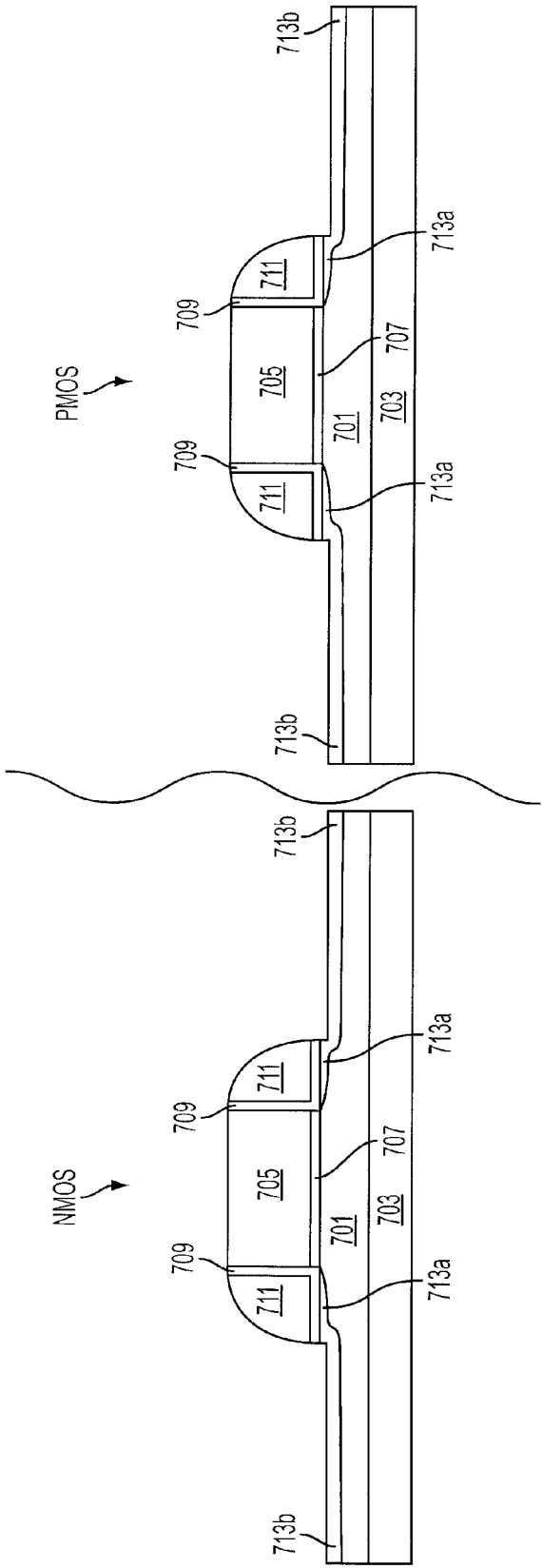


FIG. 7

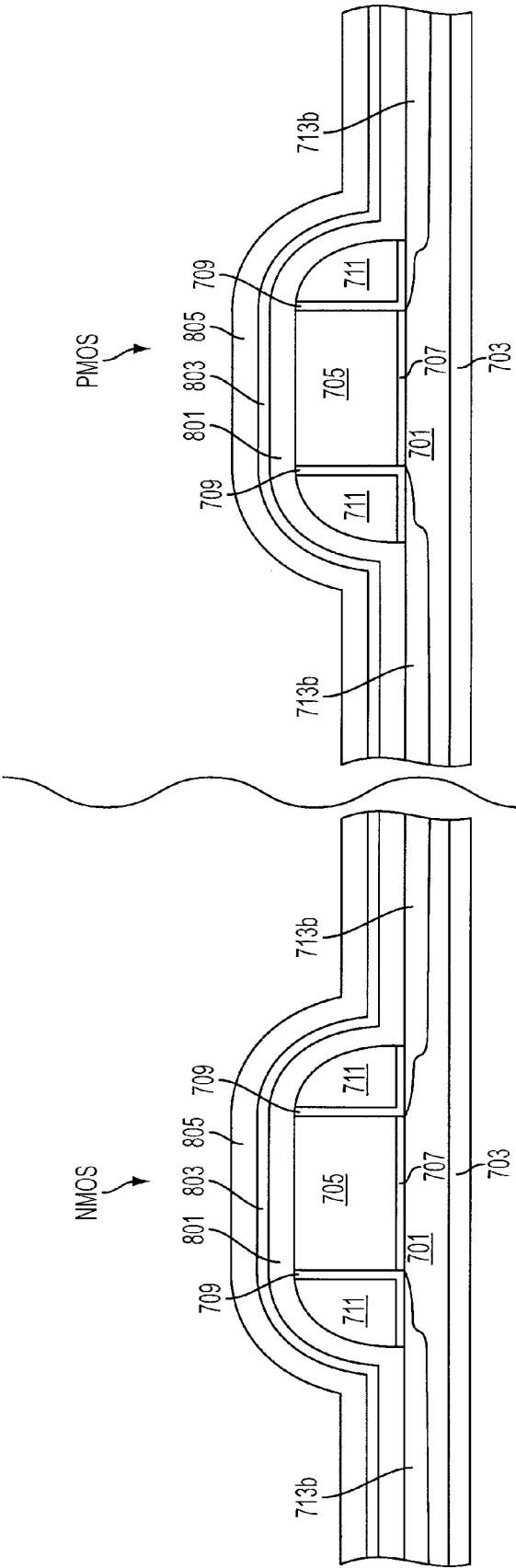


FIG. 8

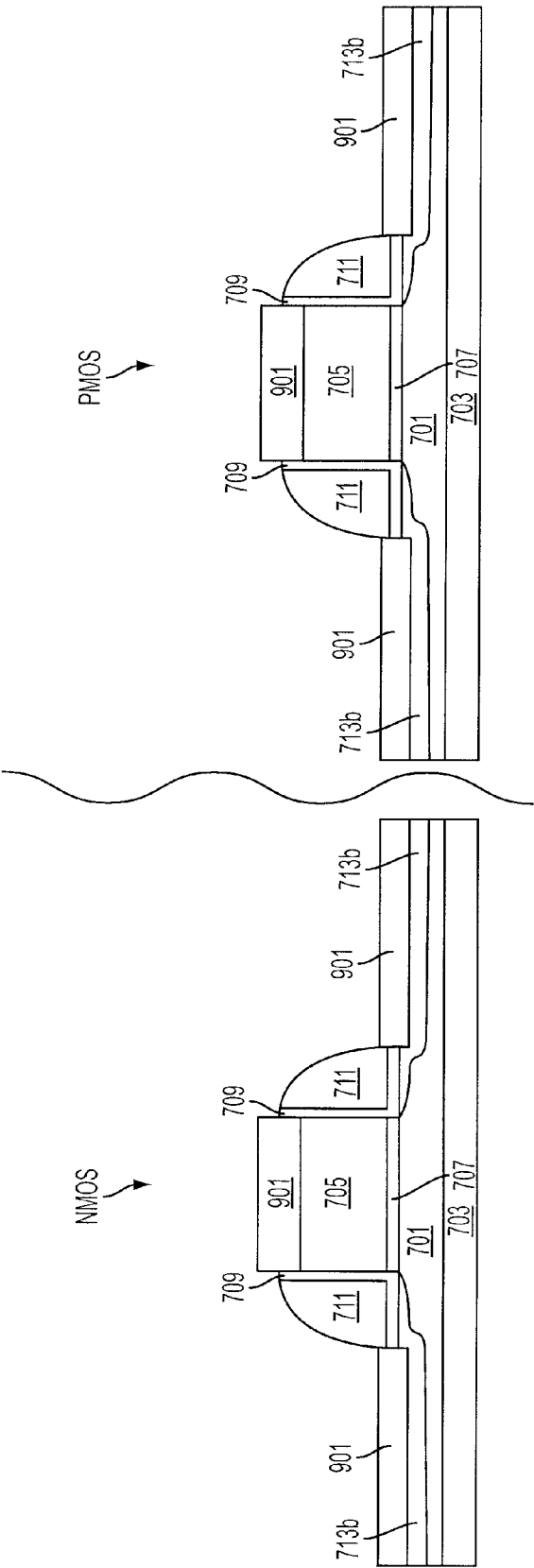


FIG. 9

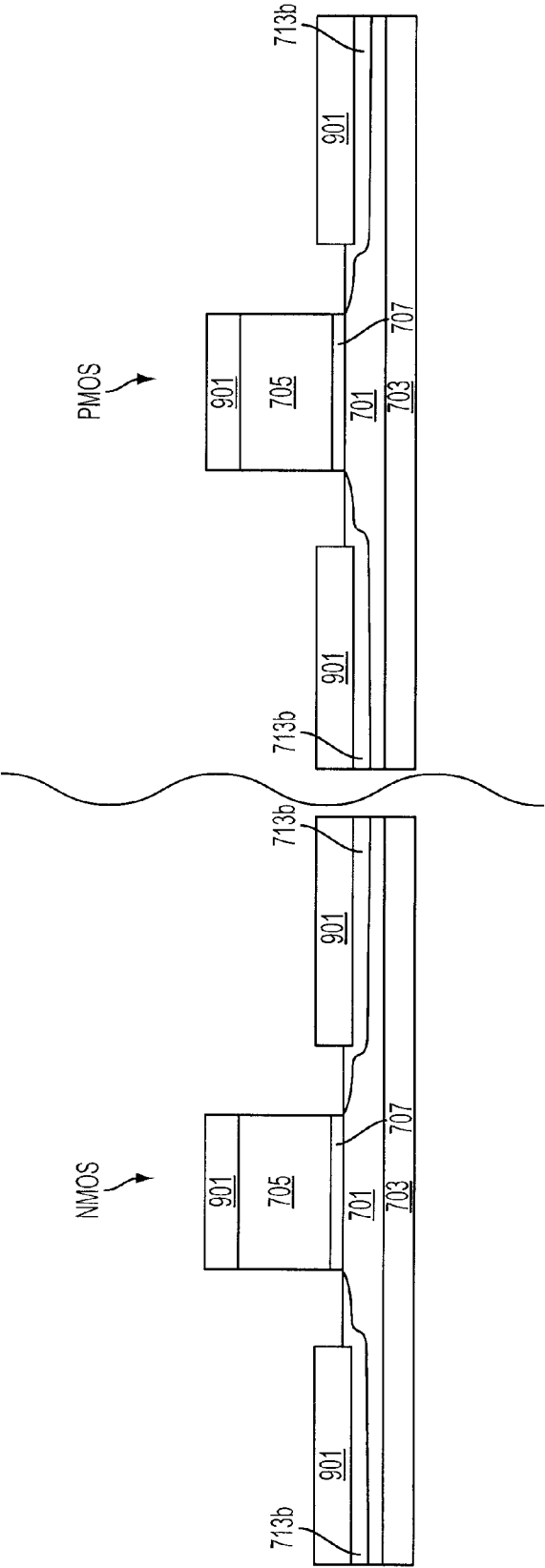


FIG. 10

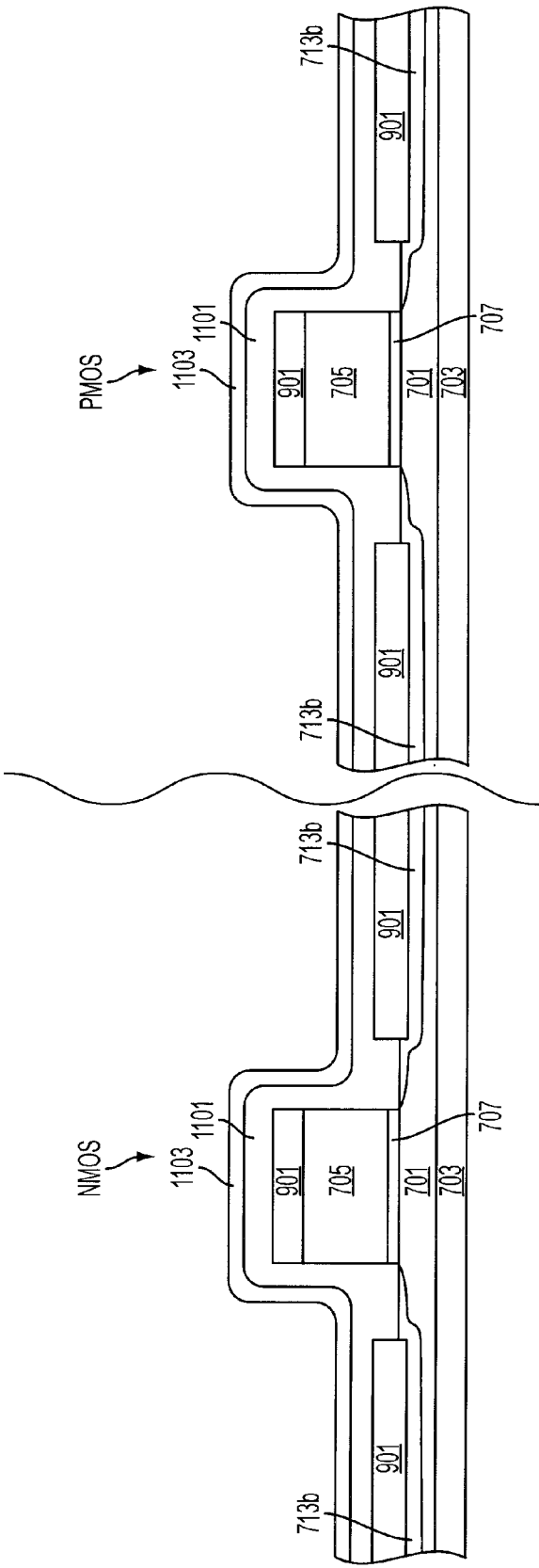


FIG. 11

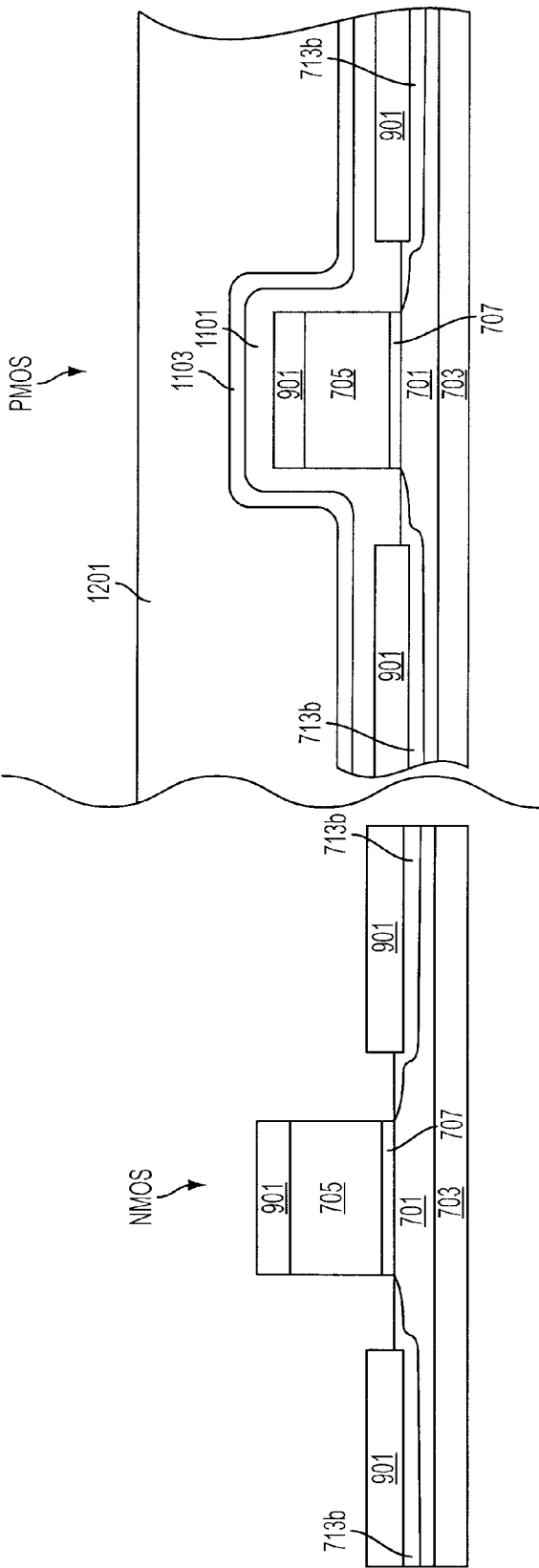


FIG. 12

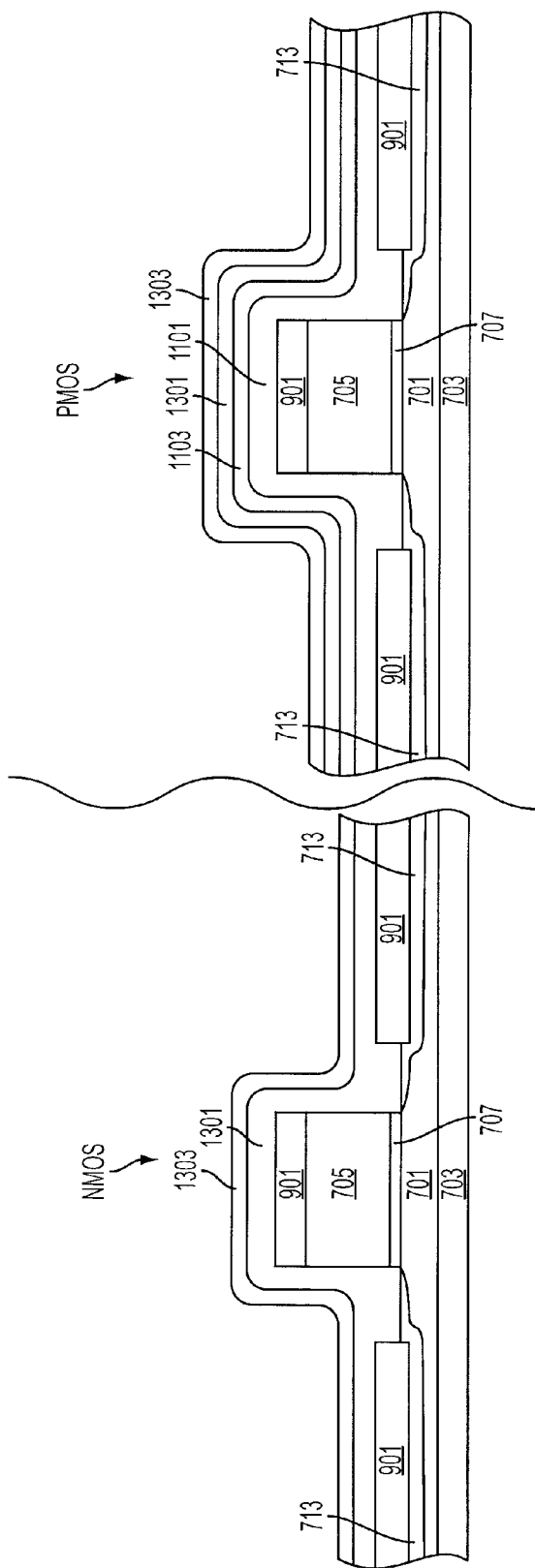


FIG. 13

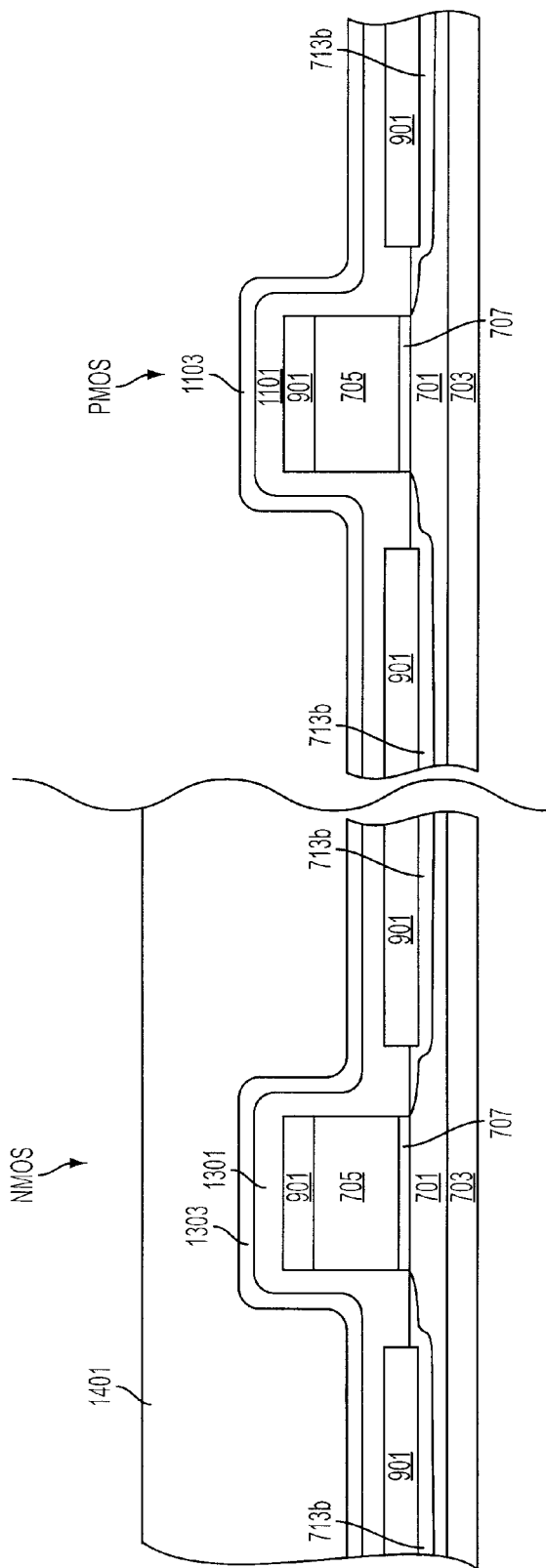


FIG. 14

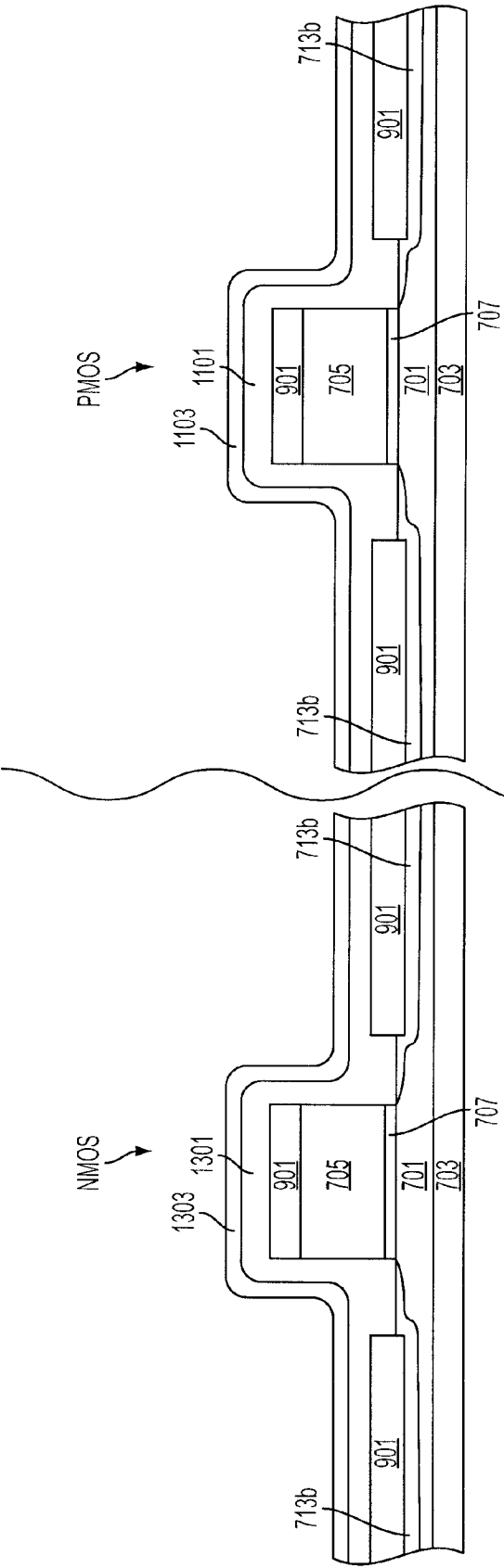


FIG. 15

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METHOD TO REDUCE MOL DAMAGE ON NISI

RELATED APPLICATIONS

This application is a Divisional application of U.S. application Ser. No. 12/366,378 filed on Feb. 5, 2009.

TECHNICAL FIELD

The present disclosure relates to semiconductor devices with improved nickel silicide layers. The present disclosure is particularly applicable in fabricating transistors with improved device performance by avoiding nickel silicide degradation.

BACKGROUND

Conventional semiconductor devices, such as the one shown in FIG. 1, include a refractory metal deposited over all exposed upper surface features of a silicon substrate **101**, preferably by means of physical vapor deposition (PVD). The most common refractory metal employed is nickel, which forms a very low resistivity silicide with silicon and does not consume as much Si as other silicides commonly employed. NiSi has a great resistance to etchants which are used to etch silicon oxide, a common insulating material used in semiconductor devices.

The refractory metal can be sputtered by DC magnetron sputtering in an ultra-high vacuum, multi-chamber system. The formation of an MOS transistor requires the formation of a gate structure **103** and source/drain junctions **105**. The gate electrode **103** typically is formed by depositing a layer of heavily doped polysilicon **103a** on a metal oxide insulating layer **103b** and etching the layers to pattern the electrode. Sidewall spacers **107** are deposited on the opposing surfaces of the patterned gate electrode with oxide liner **109** between spacer **107** and both gate electrode **103** and substrate **101**. The sidewall spacers **107** comprise silicon oxide, silicon nitride or silicon oxynitride.

Generally, in forming the MOS transistor, the refractory metal is deposited after etching the gate electrode **103** and after forming the source/drain junction **105**, as illustrated in FIG. 1. After deposition, the refractory metal layer blankets the top surface of the gate electrode **103**, sidewall spacers **107**, oxide liners **109**, and source/drain regions **105**. As a result of thermal processing, e.g., a rapid thermal annealing (RTA) process, performed in an inert or reducing atmosphere, the refractory metal reacts with underlying silicon to form electrically conductive silicide layer portions **111** on the top surface of the polysilicon gate electrode **103** and on the exposed surfaces of the substrate where source and drain regions **105** are formed. Unreacted portions of the refractory metal layer, e.g., on the silicon oxide, nitride or oxynitride sidewall spacers **107** and the silicon oxide liner **109** are then removed, as by a wet etching process selective to the metal silicide portions **111**.

Substrates based on "strained silicon" have attracted interest as a semiconductor material which provides increased speeds of electron and hole flow therethrough, thereby permitting fabrication of semiconductor devices with higher operating speeds, enhanced performance characteristics, and lower power consumption. A layer **101a** of silicon-germanium (Si—Ge) is formed on a suitable crystalline substrate, e.g., a Si wafer or a silicon-on-insulator (SOI) wafer **101b**. The Si—Ge layer has a greater lattice constant (spacing of Si and Ge atoms therein) relative to the underlying Si. As a

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consequence, the Si—Ge layer has a compressive strain. Electrons in such compressively strained layers have greater mobility than in conventional Si layers with smaller inter-atom spacings, i.e., there is less resistance to electron flow.

For example, electron flow may be up to about 70% faster compared to electron flow in conventional Si. Transistors and IC devices formed with such strained Si layers can exhibit operating speeds up to about 35% faster than those of equivalent devices formed with conventional Si, without necessity for reduction in transistor size. Conventional practices based on strained silicon technology also involve epitaxially growing a relaxed silicon layer on a tensilely stressed silicon layer which is subsequently doped to form relaxed source/drain regions in the relaxed silicon layer.

Conventional semiconductor devices typically comprise a plurality of active devices in or on a common semiconductor substrate, e.g., CMOS devices comprising at least a pair of PMOS and NMOS transistors in spaced adjacency. The mobility of electrons is faster than the mobility of holes in conventional bulk silicon substrates. Accordingly, in conventional CMOS transistors, the drive current of the PMOS transistor is less than the drive current of the NMOS transistor creating an imbalance. This imbalance is exacerbated in CMOS transistors fabricated on or within a tensilely stressed active device area formed in a strained lattice semiconductor substrate, e.g., strained Si—Ge on Si, because the increase in electron mobility is greater than the increase in hole mobility.

Channel carrier mobility of transistors based on strained Si substrates can be increased by applying a stress thereto. In forming P-channel transistors, channel carrier mobility is enhanced by applying a stressed dielectric layer exhibiting high compressive stress for increasing electron mobility. In N-channel transistors, channel carrier mobility is significantly increased by applying a stressed layer exhibiting high tensile stress for increasing hole mobility. Stressed dielectric layers, called stress liners, are applied to transistors wherein the source/drain regions are formed within the strained Si layer, and to transistors having relaxed source/drain regions formed on strained Si layers. The stress liner may comprise silicon carbide, silicon nitride or silicon oxynitride, and may be deposited by plasma enhanced chemical vapor deposition (PECVD) at a thickness of about 200 Å to about 1000 Å. Conventional PECVD conditions may be employed for deposition of a highly compressive layer or highly tensile dielectric layer. In depositing a stress liner exhibiting high compressive stress, both high frequency and low frequency power are applied. When depositing a stressed dielectric layer exhibiting high tensile stress, the low frequency power is significantly reduced.

Because the stress liners differ for P-channel and N-channel transistors, the stressed dielectric layers are typically patterned by reactive ion etching, which involves physically bombarding the surfaces to subtractively remove material. This etching process involves an "overetch" in order to clear the residue left behind. It is during this overetch process that the underlying NiSi layers are damaged, causing degradation of the NiSi. The degradation can be morphological, wherein mouse bites or missing NiSi is observed, as illustrated in FIG. 1 at **113a-113f**; or electrical, wherein the resistivity is increased. Either or both types of degradation ultimately result in poor device parameters such as higher sheet resistance and external resistance.

A need therefore exists for methodology enabling fabrication of semiconductor devices comprising transistors with degradation resistant NiSi layers and for the resulting improved semiconductor devices having reduced resistance and improved performance.

An aspect of the present disclosure is a semiconductor device comprising a transistor with reduced resistance.

Another aspect of the present disclosure is an efficient method of fabricating a semiconductor device comprising a transistor with reduced resistance.

Additional aspects and other features of the present disclosure will be set forth in the description which follows and in part will be apparent to those having ordinary skill in the art upon examination of the following or may be learned from the practice of the present disclosure. The advantages of the present disclosure may be realized and obtained as particularly pointed out in the appended claims.

According to the present disclosure, some technical effects may be achieved in part by a method comprising: forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode; forming a first layer of nickel containing a first amount of platinum over the transistor; forming a second layer of nickel containing a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; and annealing the first and second layers of nickel to form a layer of NiSi over the source/drain regions and the gate electrode, wherein the NiSi layer has a platinum composition gradient with platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

Aspects of the present disclosure include forming the first and second layers of nickel at a combined thickness of about 45 Å to about 100 Å with the first layer of nickel at a thickness of about 60% to about 80% of the combined thickness. Aspects also include annealing to form the layer of NiSi at a thickness of about 100 Å to about 220 Å. Further aspects include annealing by heating at a temperature of about 340° C. to about 450° C., such as at a temperature of about 360° C. to about 400° C. This technique can be used to effect annealing in a single step. Another aspect includes annealing by heating at a first stage at a first temperature of about 280° C. to about 350° C., for example about 300° C. to about 320° C., followed by heating at a second stage at a second temperature of about 360° C. to about 450° C., for example about 380° C. to about 420° C. Additional aspects include forming the first layer of nickel with a platinum content of about 8 weight % ("wt. %") to about 12 wt. %, for example about 10 wt. %, and the second layer of nickel with a platinum content of about 30 wt. % to about 70 wt. %, for example about 30 wt. % to about 50 wt. %. Aspects also include forming a capping layer of titanium nitride over the second layer before annealing, and removing the capping layer after annealing, followed by etching to remove unreacted nickel.

Another aspect of the present disclosure is a semiconductor device comprising: a silicon substrate; a transistor, comprising source/drain regions and a gate electrode, on the substrate; and a NiSi layer on the gate electrode and source/drain regions, the NiSi layer comprising platinum having a composition gradient, the platinum increasing in amount in a direction away from the source/drain regions and the gate electrode.

Aspects include devices with a P-channel transistor, an N-channel transistor, a first NiSi layer on the P-channel transistor, a second NiSi layer on the N-channel transistor, the first and second NiSi layers each comprising platinum having a concentration gradient with platinum increasing in amount toward the upper surface, i.e., increasing in amount in a direction away from the P-channel and N-channel transistors, a first stress liner on the P-channel transistor, and a second stress liner on the N-channel transistor. Further aspects

include the NiSi layers having a thickness of about 100 Å to about 220 Å, for example about 100 Å to about 150 Å.

Another aspect of the present disclosure is a method of fabricating a semiconductor device, the method comprising: forming a P-channel and an N-channel transistor on a silicon substrate, each transistor including a polysilicon gate electrode and source/drain regions formed in the silicon substrate; depositing a first layer of nickel containing a first amount of platinum on both transistors; depositing a second layer of nickel having a second amount of platinum on the first layer of nickel, the second amount of platinum being greater than the first amount of platinum; depositing a titanium nitride capping layer on the second layer of nickel, annealing the first and second layers of nickel to form a single layer of NiSi containing platinum having a concentration gradient with the platinum increasing in amount in a direction away from the gate electrode and source/drain regions; etching to remove the capping layer and unreacted nickel; forming a first stress liner exhibiting high compressive stress over both transistors; selectively removing the first stress liner from the N-channel transistor; forming a second stress liner exhibiting high tensile stress over both transistors; and selectively removing the second stress liner from the P-channel transistor.

Additional aspects and technical effects of the present disclosure will become readily apparent to those skilled in the art from the following detailed description wherein embodiments of the present disclosure are described simply by way of illustration of the best mode contemplated to carry out the present disclosure. As will be realized, the present disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects, all without departing from the present disclosure. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawing and in which like reference numerals refer to similar elements and in which:

FIG. 1 schematically illustrates a conventional transistor device with a degraded NiSi layer;

FIGS. 2 through 6 schematically illustrate sequential steps of a method in accordance with an embodiment of the present disclosure;

FIGS. 7 through 15 schematically illustrate sequential steps of a method in accordance with another embodiment of the present disclosure.

DETAILED DESCRIPTION

In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of exemplary embodiments. It should be apparent, however, that exemplary embodiments may be practiced without these specific details or with an equivalent arrangement. In other instances, well-known structures and devices are shown in block diagram form in order to avoid unnecessarily obscuring exemplary embodiments.

The present disclosure addresses and solves the problem of NiSi layer degradation during subsequent processing, as during middle of the line processing to remove overlying stress liners, which NiSi layer degradation causes resistance problems and adversely affects transistor performance. In accordance with embodiments of the present disclosure, a transis-

tor device is provided having a NiSi layer formulated to resist degradation, advantageously using methodology that does not impact process flow. In accordance with embodiments of the present disclosure, NiSi layer damage is prevented or significantly reduced by formulating the NiSi layer with a platinum composition gradient with the amount of platinum increasing in a direction toward its upper surface, i.e., in a direction away from the source/drain regions and the gate electrode, thereby shielding the NiSi layer from physical damage during post NiSi processing as when removing an overlying stress liner by etching.

Embodiments of the present disclosure include forming two layers of nickel on a silicon substrate and a silicon gate electrode, the first layer having a relatively low concentration of platinum and the second having a relatively high concentration of platinum, and annealing the two layers to form a single layer of NiSi. During annealing, platinum diffuses to form a composition gradient with the amount of platinum increasing in a direction away from the substrate and gate electrode. It was found that such a concentration gradient of platinum effectively shields the NiSi layer from physical damage during subsequent processing, e.g., etching, while concomitantly maintaining the integrity of the NiSi layer.

Methodology in accordance with embodiments of the present disclosure includes forming a transistor on a substrate, the transistor including source/drain regions and a gate electrode, depositing a first layer of nickel containing a relatively low amount of platinum, for example about 8 wt. % to about 12 wt. %, e.g., about 10 wt. %, over the transistor, forming a second layer of nickel containing a relatively high amount of platinum, such as about 30 wt. % to about 70 wt. %, e.g., about 30 wt. % to about 50 wt. %, on the first layer. The first and second layers of nickel are formed at a combined thickness of about 45 Å to about 100 Å, with the first layer of nickel at a thickness of about 60% to about 80% of the combined thickness. A capping layer of silicon nitride is formed over the second nickel layer. An RTA process is then implemented to convert the platinum containing nickel layers into a single layer of NiSi over the source/drain regions and the gate electrode, such that the NiSi layer has a platinum composition gradient with the amount of platinum increasing in a direction away from the source/drain regions and the gate electrode. The resulting platinum concentration gradient protects the NiSi film without increasing the resistivity of the layer.

The annealing process may be controlled to form a layer of NiSi at a thickness of about 100 Å to about 220 Å. Annealing may comprise a single heating stage at a temperature of about 340° C. to about 450° C., such as at a temperature of about 360° C. to about 400° C. Alternatively, a two stage annealing process can be performed, as by heating at a first temperature of about 280° C. to about 350° C., such as about 300° C. to about 320° C., followed by heating at a second temperature of about 360° C. to about 450° C., such as about 380° C. to about 420° C. In accordance with embodiments of the present disclosure, the capping layer is removed after annealing followed by etching to remove unreacted nickel, employing conventional methodology.

Embodiments of the present disclosure include forming two transistors, one P-channel and one N-channel transistor. A first stress liner exhibiting high compressive stress can be formed over both transistors, and the first stress liner is selectively removed, as by etching, from the N-channel transistor. A second stress liner exhibiting high tensile stress can be subsequently formed over both transistors, and the second stress liner is selectively removed, as by etching, from the P-channel transistor. During etching of the stress liners, the

thus formed platinum concentration gradient prevents the NiSi from being etched and damaged.

Still other aspects, features, and technical effects will be readily apparent to those skilled in this art from the following detailed description, wherein preferred embodiments are shown and described, simply by way of illustration of the best mode contemplated. The disclosure is capable of other and different embodiments, and its several details are capable of modifications in various obvious respects. Accordingly, the drawings and description are to be regarded as illustrative in nature, and not as restrictive.

A method in accordance with an embodiment of the present disclosure is schematically illustrated in FIGS. 2 through 5, wherein like reference characters denote like features. Adverting to FIG. 2, strained Ge—Si layer 201 is formed over Si layer 203. Ge—Si layer 201 may be globally strained or locally strained under the source/drain regions. Gate electrode 205 is formed over strained Ge—Si layer 201 with a gate dielectric 207 therebetween. Lightly doped source/drain regions 213a are formed in Ge—Si layer 201 using the gate electrode 205 as a mask in a conventional manner. A sidewall spacer comprising a silicon oxide liner 209, as at a thickness of about 60 Å to about 600 Å, is formed on side surfaces of the gate electrode 205 and on a portion of the upper surface of strained Ge—Si layer 201. Silicon nitride spacers 211 are then formed on the silicon oxide liner 209 as by employing PECVD followed by etching.

It should be appreciated that liner 209 may be deposited by ALD and may also comprise silicon nitride. Silicon oxide liner 209 advantageously prevents consumption of the gate electrode 205 by silicidation on the side surfaces thereof, and advantageously prevents a subsequently formed thin layer of NiSi on the silicon nitride sidewall spacers 211 from contacting a NiSi contact layer on the upper surface of the gate electrode 205 and/or from contacting NiSi contact layers on the upper surface of the strained Si layer 201, thereby preventing NiSi bridging along the silicon nitride sidewall spacers.

Subsequently, heavily doped source/drain regions 213b are formed in Ge—Si layer 201 using gate electrode 205 and sidewall spacers 209 and 211 as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

Adverting to FIG. 3, a first layer of nickel 301 is deposited over the source/drain regions 213b, the spacers 211, the oxide liner 209, and the gate electrode 205. First layer of nickel 301 may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. The first layer of nickel can be formed by sputtering using a target comprising Ni and 10 wt. % Pt. Over the first layer of nickel 301, a second layer of nickel 303 is deposited. The second layer of nickel 303 may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The second layer of nickel can be sputtered using a target of Ni containing an amount of Pt within the range of 30 to 100%. The first nickel layer 301 and second nickel layer 303 may be formed to a combined thickness of about 45 Å to about 100 Å using a conventional deposition tool, for example on an AMAT Endura sputtering system, at a power of 1500 W to 2500 W, e.g., 2000 W, and using an Argon flow of 20 sccm to 40 sccm, e.g., 2000 sccm, or an Argon and Hydrogen mixture with a flow of 3 sccm to 10 sccm, e.g., 5 sccm. The first layer of nickel 301 may be formed at a thickness of about 60% to about 80% of the combined thickness, such as about 70% of the combined thickness.

A capping layer **305** may be formed over the second nickel layer **303** to prevent oxidation of the underlying metal. Capping layer **305** may be titanium nitride.

RTA and etch processes are performed to form a NiSi layer **401** over the source/drain regions **213b** and the gate electrode **205**, as illustrated in FIG. 4. The resulting NiSi layer **401** has a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the silicon interface **403** to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layer **401** may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å.

Nickel silicide is commonly doped with platinum to improve the material and electrical properties. Platinum has proven to be effective in preventing the nickel disilicide formation (NiSi₂). It was found that strategic platinum doping can also be used to effectively shield a NiSi layer from physical damage during MOL processing (post NiSi processing) while concomitantly maintaining the NiSi integrity. However, platinum also increases the resistivity of NiSi. In accordance with embodiments of the present disclosure, effective shielding of a NiSi layer is achieved while avoiding a problematic increase in resistivity by compositionally grading a platinum doped NiSi layer.

The RTA process may be a one-step or two-step process (with a clean in between the two steps) in an inert or reducing atmosphere, e.g., a nitrogen (N₂)-containing atmosphere. The single step anneal may be implemented at a temperature of about 340° C. to about 450° C., for example at a temperature of about 360° C. to about 400° C., for about 5 to 35 seconds. During the single step anneal the nickel layers are converted directly to a low resistivity monosilicide (NiSi).

The two-step anneal may be conducted by first heating at a temperature of about 280° C. to about 350° C., for example about 300° C. to about 320° C., e.g., 300° C., during which metal rich silicide, Ni₂Si, is formed. During the second step, the temperature may be raised to about 360° C. to about 450° C., such as about 380° C. to about 420° C., e.g., 420° C. During the second annealing step, the metal rich silicide is converted to monosilicide. The total time for the two step annealing can be the same as for the single step anneal process, for example 5 seconds for the first step and 30 seconds for the second step. The two-step anneal therefore reduces the time at which annealing occurs at a high temperature, thereby reducing the rough interfaces between the NiSi and the underlying Si layer.

After NiSi layer **401** is formed, the capping layer **305** and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are removed by an anisotropic sputter etching or by immersing the wafer into a solution of H₂SO₄, H₂O₂, and water (SPM) or a solution of NH₄OH, H₂O₂ and water (APM).

As illustrated in FIG. 5, the liner **209** and sidewall spacers **211** are then removed, exposing a portion of the upper surface of strained Si layer **201** between the silicide layers **401** and the side surfaces of gate electrode **205**, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer. A highly stressed dielectric layer **601**, such as a silicon nitride layer exhibiting high compressive stress, is then deposited by PECVD, as shown in FIG. 6. The highly stressed dielectric layer **601** serves to increase channel electron mobility, thereby increasing the drive current.

Another embodiment of the present disclosure is schematically illustrated in FIGS. 7 through 15. Adverting to FIG. 7, a CMOS device is schematically illustrated comprising an NMOS transistor portion at the left and a PMOS transistor

portion on the right, wherein similar features are denoted by similar reference characters. Strained Si—Ge layer **701** is formed over Si layer **703**. As in previous discussed embodiments, Si—Ge layer **701** may be globally strained or locally strained in the source/drain regions. Gate electrodes **705** are formed over strained Si—Ge layer **701** with gate dielectric **707** therebetween. Lightly doped source/drain regions **713a** are formed in Si—Ge layer **701** using gate electrodes **705** as a mask in a conventional manner. Sidewall spacers comprising oxide liner **709**, as at a thickness of about 60 Å to about 600 Å, are formed on side surfaces of gate electrodes **705** and on portions of the upper surface of strained Si—Ge layer **701**. Silicon oxide liner **709** can be formed in the same manner as discussed with respect to silicon oxide liner **209** in FIG. 2. Silicon nitride spacers **711** are then formed on silicon oxide liners **709** as by employing PECVD followed by etching.

Subsequently, heavily doped source/drain regions **713b** are formed in Si—Ge layer **701** using gate electrodes **705** and sidewall spacers **709** and **711** as a mask, employing conventional implantation techniques, as by ion implanting boron, phosphorus, or arsenic.

As illustrated in FIG. 8, a first layer of nickel **801** is deposited over the source/drain regions **713b**, spacers **711**, oxide liners **709**, and gate electrodes **705**. First layer of nickel **801** may be formed, for example, with a platinum content of about 8 wt. % to about 12 wt. %, such as about 10 wt. %. Over the first layer of nickel **801**, a second layer of nickel **803** is deposited. Second layer of nickel **803** may be formed, for example, with a platinum content of about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %. The first nickel layer **801** and second nickel layer **803** may be formed in a conventional deposition environment, for example on an AMAT Endura sputtering system, to a combined thickness of about 45 Å to about 100 Å. The first layer of nickel **801** may be formed at a thickness of about 60% to about 80% of the combined thickness, such as about 70% of the combined thickness.

A capping layer **805** is formed over the second nickel layer **803** to prevent oxidation of the underlying metal. Capping layer **805** may be titanium nitride.

RTA and etch processes are performed to form NiSi layers **901** over source/drain regions **713b** and gate electrodes **705**, as illustrated in FIG. 9. The resulting NiSi layers **901** have a composition gradient of platinum from about 8 wt. % to about 12 wt. %, such as about 10 wt. %, at the bottom surface that contacts the silicon gate electrodes **705** and source/drain regions **713b**, to about 30 wt. % to about 70 wt. %, such as about 30 wt. % to about 50 wt. %, at the opposite surface. NiSi layers **901** may be formed at a thickness of about 100 Å to about 220 Å, such as at a thickness of about 100 Å to about 150 Å. The RTA process may be a one-step or two-step process, as described above with respect to FIG. 4.

After NiSi layers **901** are formed, the capping layer **805** and unreacted Ni on the wafer are then easily removed, as by wet chemical stripping. In an embodiment of the present disclosure, the capping layer and unreacted Ni are removed by an anisotropic sputter etching or by immersing the wafer into a solution of H₂SO₄, H₂O₂, and water (SPM) or a solution of NH₄OH, H₂O₂ and water (APM).

As illustrated in FIG. 10, liners **709** and sidewall spacers **711** are then removed exposing portions of the upper surface of strained Si layer **701** between the silicide layers **901** and the side surfaces of gate electrodes **705**, with a very thin oxide layer (not shown) thereon, e.g., less than 50 Å, serving as a buffer layer.

A highly compressive stressed silicon nitride film **1101**, e.g., having a compressive stress greater than 1.5 GPa, is then

deposited over both the NMOS and PMOS transistors as shown in FIG. 11. Deposition of highly compressive stressed silicon nitride film 1101 may be implemented in a conventional manner. Multiple layer deposition and plasma treatment further increases compressive stress. Subsequently, a thin oxide or oxynitride film 1103 is deposited, as illustrated in FIG. 11, by a conventional CVD process. Typically oxide or oxynitride film 1103 is deposited at a thickness of about 30 Å to about 60 Å.

Subsequently, a mask 1201, e.g., photoresist or hard mask is applied over the PMOS transistor, as illustrated in FIG. 12, and the oxide or oxynitride film 1103 and high compressive stress silicon nitride film 1101 are removed from the NMOS transistor.

Adverting to FIG. 13, the mask 1201 is removed from the PMOS transistor. A highly tensile stressed silicon nitride film 1301, having a tensile stress of greater than 1.5 GPa, is then deposited over both the PMOS and NMOS transistors. Deposition of highly tensile stressed film 1301 can be implemented in a conventional manner. A thin oxide or oxynitride film 1303 is then deposited by a conventional CVD process, as at a thickness of about 30 Å to about 60 Å.

Adverting to FIG. 14, a mask 1401, such as a photoresist or hard mask, is then applied over the NMOS transistor, and the oxide or oxynitride film 1303 and the highly tensile stressed silicon nitride film 1301 are selectively removed from the PMOS transistor stopping on the oxide or oxynitride film 1103. The mask 1401 is then removed, and the resulting structure is illustrated in FIG. 15 which comprises an oxide or oxynitride film 1303 and a highly tensile stressed silicon nitride film 1301 over the NMOS transistor and an oxide or oxynitride film 1103 and a highly compressive stressed silicon nitride film 1101 over the PMOS transistor. The resulting CMOS device comprises both PMOS and NMOS transistors with increased channel carrier mobility and, hence, increased drive current, without increasing the resistivity of the transistors.

The embodiments of the present disclosure can achieve several technical effects, including resistance reduction, improved transistor and circuit performance, and reduced power requirements. The present disclosure enjoys industrial applicability in any of various types of highly integrated semiconductor devices.

In the preceding description, the present disclosure is described with reference to specifically exemplary embodiments thereof. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the present disclosure, as set forth in the claims. The specification and drawings are, accordingly, to be regarded as illustrative and not as restrictive. It is understood that the present disclosure is capable of using various other combinations and embodiments and is capable of any changes or modifications within the scope of the inventive concept as expressed herein.

What is claimed is:

1. A semiconductor device comprising:

a silicon substrate,

a transistor, comprising source/drain regions and a gate electrode, on the substrate; and

a nickel silicide layer on the gate electrode and source/drain regions,

wherein the nickel silicide layer comprises platinum having a composition gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direc-

tion away from the source/drain regions and the gate electrode, from between about 8 wt. % and about 12 wt. % at the silicon interface.

2. The semiconductor device according to claim 1, wherein the concentration gradient of platinum in the nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

3. The semiconductor device according to claim 2, wherein the concentration gradient of platinum in the nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

4. The semiconductor device according to claim 1, wherein the nickel silicide layer has a thickness of about 100 Å to about 220 Å.

5. The semiconductor device according to claim 1, wherein the nickel silicide layer has a thickness of about 100 Å to about 150 Å.

6. The semiconductor device according to claim 1, wherein the silicon substrate comprises a strained silicon germanium (Si—Ge) layer on a silicon layer.

7. The semiconductor device according to claim 1, further comprising a stress liner on the nickel silicide layer.

8. The semiconductor device according to claim 7, further comprising an oxide or oxynitride layer on the stress liner.

9. The semiconductor device according to claim 8, wherein the stress liner comprises stressed silicon nitride.

10. A semiconductor device comprising:

a silicon substrate;

a P-channel transistor and an N-channel transistor, each comprising source/drain regions and a gate electrode, on the silicon substrate; and

a first nickel silicide layer on the gate electrode and source/drain regions of the P-channel transistor; and

a second nickel silicide layer on the gate electrode and source/drain regions of the N-channel transistor,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount in a direction away from the P-channel and N-channel transistors, from between about 8 wt. % and about 12 wt. % at the silicon interface.

11. The semiconductor device according to claim 10, further comprising:

a first stress liner on the P-channel transistor; and

a second stress liner on the N-channel transistor.

12. The semiconductor device according to claim 11, wherein the first stress liner exhibits high compressive stress and the second stress liner exhibits high tensile stress.

13. The semiconductor device according to claim 12, wherein the concentration gradient of platinum in each nickel silicide layer ranges from between about 8 wt. % and about 12 wt. % to between about 30 wt. % and about 100 wt. %.

14. The semiconductor device according to claim 13, wherein the concentration gradient of platinum in each nickel silicide layer ranges from about 10 wt. % to between about 30 wt. % and about 50 wt. %.

15. The semiconductor device according to claim 12, wherein each nickel silicide layer has a thickness of about 100 Å to about 220 Å.

16. The semiconductor device according to claim 15, wherein each nickel silicide layer has a thickness of about 100 Å to about 150 Å.

17. The semiconductor device according to claim 12, wherein the silicon substrate comprises a strained silicon germanium (Si—Ge) layer on a silicon layer.

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18. The semiconductor device according to claim 12, further comprising an oxide or oxynitride layer on the first and second stress liners.

19. The semiconductor device according to claim 18, wherein the first and second stress liners comprise stressed silicon nitride. 5

20. A semiconductor device comprising:

a silicon substrate;

a strained silicon germanium (Si—Ge) layer on the silicon substrate; 10

a P-channel transistor comprising source/drain regions and a gate electrode on the strained Si—Ge layer;

an N-channel transistor comprising source/drain regions and a gate electrode on the strained Si—Ge layer, spaced from the P-channel transistor; 15

a first nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the P-channel transistor;

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a second nickel silicide layer, having a thickness of about 100 Å to about 220 Å, on the gate electrode and source/drain regions of the N-channel transistor;

a compressively stressed silicon nitride layer on the first nickel silicide layer;

a tensile stressed silicon nitride layer on the second nickel silicide layer;

a first oxide or oxynitride layer on the compressively stressed silicon nitride layer; and

a second oxide or oxynitride layer on the tensile stressed silicon nitride layer,

wherein each of the first and second nickel silicide layers comprises platinum having a concentration gradient, from a nickel silicide surface at a silicon interface to an opposite surface of the nickel silicide, with platinum increasing in amount from between about 8 wt. % and about 12 wt. % at the silicon interface to between about 30 wt. % and about 100 wt. % in a direction away from the P-channel and N-channel transistors.

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